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ESOL Nexus

ESOL Nexus professional development

Teaching mathematics to ESOL learners

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Introduction

ESOL Nexus professional development modules are intended to help teachers to develop in their role in a range of contexts. They have been written by expert practitioners and quality assured by a wider team. We hope that you will find this module useful. For other professional development opportunities, see

http://esol.britishcouncil.org/content/teachers/staff-room/continuing-professional-development.

This module is about teaching mathematics to ESOL learners. It aims to raise your awareness of the challenges and differences that learners might meet whilst learning mathematics in English, so that you can address them. This will enable learners to be successful in their study of mathematics, and improve their English skills. It is relevant for learners at any level of mathematics and English as it is focused on the underpinning knowledge required for success.

Using this module

You can either work through the module at your own pace, or join with others to look at the content together.

If you are reading the module on a computer you can type your reflections into the boxes provided, and save your progress for the next time you return to the module.

There is a glossary at the end of the module, which explains any words that might be unfamiliar.

Contents

Introduction to number

This step will help you to answer the following questions:

- How is the numbering system in English different from other languages?
- What challenges will learners face when rounding and estimating in maths problems?
- How can the written form of numbers operations in English impact on understanding?

There are many challenges that ESOL learners face in maths classes, whether they have some maths knowledge in another language or not. Mathematics is often about patterns; about utilising knowledge gained in one area and applying it in another, but it will not be possible to do this unless the building blocks that underpin the system are firmly in place.

One of the most fundamental of these building blocks is number. Whilst most of the world's populations follow the same number system there are some anomalies in English that could impact on understanding. These include both ways of writing numbers, and their pronunciation. For instance, in some languages the use of commas and full stops may be reversed and word endings may not be pronounced, all impacting on the ability of the learner to understand and be understood.

Many learners may not be familiar with Romanised script or writing on a page from left to right which could impact on the speed of progress that learners can make. This section aims to raise your awareness of how challenging maths in English can be for learners and what barriers certain groups of learners might face.

Rounding and estimation are both concepts that can trouble ESOL leaners. Firstly, ‘round’ has a number of meanings in English, of which the mathematical version (and there are two) is at the higher end of English learning. Secondly, estimation in mathematics does not seem to exist as an idea in all other languages; learners may ask you why they should estimate, when they can work the answer out exactly. In this section you will explore ideas on delivery so that you can enable ESOL learners to engage with the curriculum.

In English there are many different ways of phrasing questions around the basic number operations: add, subtract, multiply and divide. This can be a challenge for any learner, but especially those whose first language is not English.

You will explore how many ways there are of asking what is basically the same question, and how important it is that learners accurately decode word problems in order to be successful in maths.

Lastly, the maths level at which the various skills are required is in brackets after the skill first appears, for instance ‘Rounding a number to the nearest 10 (E2/N2)’ shows that this skill is needed at Entry 2/National 2 and above.

Writing numbers

Written numbers are frequently needed in everyday life (E1/N1- up to 10; E2/N2- up to 100; E3/N3 up to 1,000; L1/L2/N4/5 any number): they are used when writing cheques, when confirming amounts in both number and written form, and interpreting prices.

English spelling can raise issues for learners: consider won and one, for and four, or ate and eight. These words have different meanings, are spelt differently but sound identical. Note that forty does not follow the same spelling pattern as four.

Pronunciation of numbers can be a challenge for learners. For instance the pronunciation of v and w can be reversed, the th sound may not be present and word endings may not usually be heard in other languages.

Numbers follow patterns, but these patterns can be interrupted: for tens we say the number followed by teen, but this does not apply to one, two, three or five. In English 'and' is inserted after hundred, or if the hundred is not there, for instance in one hundred and ten or two thousand and four.

The way numbers are written to show place value can differ. In English 100,006 is one hundred thousand and six, but to many learners this number looks like a decimal number: one hundred point zero, zero, six.

When writing money (E2/N2) in English a full stop is used between pounds and pence: £2.15 is two pounds, fifteen pence. In many other languages the demarcation between large and small denominations is

Step 1a

written with a comma: two euros, fifteen cents could be written as €2,15 or 2,15€.

Other languages can have special words for place value that are absent in English, such as lekh for one hundred thousand in Hindi. The repetition of hundred in a string of numbers can confuse some learners.

Step 1a

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Activity 1

Write each number in words.

44 ....

107 ....

506,508 ....

....

Why might learners find any of these numbers challenging to recall?

....

....

....

Activity 2

Write this amount of money in numbers:

Twenty three pounds, sixty eight pence

....

What mistakes might learners make?

....

....

....

Step 1a

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Rounding and Estimation

Rounding

Rounding is often used in real life, for instance if discussing distances between towns and cities or when shopping: we might say a CD costs ten pounds, when it is actually £9.99 (E3/N3).

The 'rounding rule' says numbers should be rounded up when the last digit is five or more and rounded down when it is 4 or less, but this may not always apply, as in the example of rounding to the nearest five.

Round has a number of different meanings which will need to be clear to learners. Ask learners to pick out shapes which are round or curved from a selection of 2D shapes. Learners could then identify how many shapes there are, and be shown how to round that number to, say, the nearest ten (E2/N2). Learners could use till receipts to round each cost to the nearest pound.

Estimation

Learners need to know that there are a number of words that might be used in place of 'estimate', such as 'about', 'roughly' or 'approximately' (E3/N3). Show examples of questions using different vocabulary, with suggested answers.

Examples of real life estimations help learners see its relevance: viewing figures for TV programmes are estimates: a sample of the population is taken, and extrapolated up to give an estimate.

Newspapers are a useful source of estimations. Ask learners to highlight and discuss examples taken from a local paper.

Estimation appears in the Curriculum at E2/N2: learners may be asked to estimate a length, weight or capacity, and there may be more than one correct answer.

Ask learners for rough answers to questions regularly. Learners from cultures where accuracy is highly valued may struggle to be comfortable with the idea of estimation, so it is important to reinforce this concept.

Activity 3

Round this number to the nearest 5:

- 12.49

....

Rephrase this question using money to make it more meaningful for ESOL learners:

When learners can round accurately and successfully they are ready to move on to estimation.

....

....

Activity 4

There are three adult classes at college. The registers show that there are 24, 17 and 19 learners in each class. If about 80% of students are expected to attend, along with 4 members of staff approximately how many buses are needed for a student outing? Each bus holds 48 people.

Which key word in this question informs learners that an estimate is required?

....

Step 1b

Vocabulary for Number Operations

There are four number operations: addition and subtraction (E1/N1), multiplication (E2/N2) and division (E3/N3). Each operation has its own symbol: +, -, x and ÷, and many words that could be used to indicate which operation is required. At the end of sums we find the equals sign, =. This indicates that whatever is on one side of the sum, or equation, is exactly the same as what is on the other side; if it is not exactly the same, the equals sign should not be used.

Addition is what is known as a commutative activity, which means that the numbers can be put in any order; the answer will be the same. For instance the answer to 45 + 71 + 89 will be the same as 89 + 71 + 45. Multiplication is also commutative, but subtraction and division are not: 89 – 17 does not equal 17 – 89, and 45 ÷ 5 is not the same as 5 ÷ 45, although they are both perfectly viable sums. Subtraction and division are known as non-commutative.

Here are some examples of the different vocabulary that might be used in word problems involving different number operations:

- Addition: add, more than, total, sum, altogether, and, plus, increase
  Example: What is 5 and 4 altogether?
- Subtraction: subtract, minus, deduct, less than, difference between, decrease
  Example: What is the difference between 46 and 79?
- Multiplication: Multiply, lots of, times, product of
  Example: What is the product of 16 and 7?

Step 1c

- Division: divide, share, go into

  Example: Share 30 oranges out between six people.

Note it is not possible to know what number operation is required unless the language is understood, and one of the above questions does not even contain a question mark.

Activity 5

What other meanings do the words sum and total have that might confuse ESOL learners?

Activity 6

Look at these two sentences:

1) What is five less than 92?
2) What is 1,000 minus 650?

Note that the number we are taking away from can appear in the first or second place in the question.

What mistake might learners make?

Step 1c

Fractions, Decimals, Percentages, Ratio and Proportion

This step will help you to answer the following questions:

- Which fractions have non-regular names?
- Why might learners make a mistake when finding 30% off £5?
- Why might ratio and proportion questions place particular demands on ESOL learners?

Fractions, decimals and percentages can be seen as different ways of writing the same value; for instance a half is the same as 0.5, which is the same as 50 per cent. Learners are often unclear about when and how to use each one, but broadly, fractions are used to describe parts of something, decimals can be used in money, measuring, or to describe probability, and percentages are often used to show discounts when shopping. In English, decimal and fraction numbers can be used interchangeably; for instance we might see 0.25, but still say 'a quarter'.

A small number of fractions have non-regular names, namely a half, a third and a quarter. After that they follow the pattern of adding 'th' after the number, as in one sixth, but one fifth and one twelfth are also exceptions. This corresponds with the place value after the decimal point: 0.1 is 'one tenth', but it could be read as 'nought point one' and learners will need to know both forms.

Finding a half or a quarter of quantities, a small number of items, or shapes appears as early as E2/N2 in the English and Scottish curricula, as does reading, writing and understanding decimal numbers up to 2 decimal points. This does not usually present a challenge if learners already have

Step 2

money skills, although they will need to be aware of the use of a point rather than a comma when writing money (see previous section).

Percentages, ratio and proportion all come into curricula at L1/N4, but it could benefit learners to have some knowledge of these from a far lower level, as they appear so much in real life. For instance in sales, reductions are often given in percentage, such as ‘25% off the normal price’. This is clearly very different from ‘25% of the normal price’, and we will cover this in more depth in Step 3.

The spoken word order for fractions and percentages is not the same in all languages, so some groups of learners may need to practise this more than others.

Ratio often appears on packaging, for instance to inform consumers how much water should be used when making up squash.

Proportion is used when cooking if, for instance, a recipe is for 4 people, but enough is needed for, say, two people. In this case learners will need to halve the amounts required, bringing us back to fractions.

Questions at all levels are likely to be contextualised, but some contexts may be outside of ESOL learners knowledge and/or comfort zone, such as those based around alcohol (mixing cocktails), or gambling (using dice or cards). These types of questions should not however appear on exam papers (Ref: Fair Access by Design).

Fractions and Decimals

A collection of coloured objects can be very useful when starting to teach fractions. For instance you may have a selection of pens, pencils or crayons in the classroom. Ask learners to count how many there are in total, then how many are, say, red? What fraction is this of the total amount? Then ask learners what fraction is blue?

Learners can be asked to cut out or be given paper shapes, such as circles, rectangles or squares. Show and/or ask learners to divide them into halves, quarters or other common fractions (E2/N2), by measuring, colouring and labelling.

Learner will need to know at E3/N3 that a fraction is not a proper fraction unless it is in its lowest form or lowest terms. Matching activities which show various equivalent fractions can be used in a number of ways, including asking learners to group them and show the lowest form. For instance two quarters is the same as one half; two quarters is one half in its lowest form.

Decimals appear at E2/N2, usually in the context of money. Learners will be expected to be able to find out a cost, and work out how much change is needed from specified denominations of currency. Role play with an imaginary shop and UK currency can help learners to become proficient in handling money and checking change.

Rulers and measuring tapes can be used for learners to practise using decimal numbers in context. For instance a pen could be 13.5 centimetres long. Ask learners to choose three objects in the room and measure them, using decimals in their answers. Can higher level learners convert those decimals to fractions?

The use of decimal numbers to express the likelihood of an occurrence comes in probability (L1/N4).

Activity 7

What is three fifths of £1?

What mistake might learners make when writing the answer to this question?

Activity 8

Place value after the decimal point:
0.001 is one thousandth

What other way can this decimal number be said?
Why might ESOL learners struggle to pronounce 'thousandths'?

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Percentages

Percentages are used primarily when shopping, especially in the sales, or when making comparisons, such as on rates of interest between different bank accounts. The work on percentages will first be examined at L1/N4, and include percentage increases and decreases, as well as finding percentage parts of quantities and measurements, but it is such a common skill it can be worth looking at with ESOL learners even if they are at lower levels with Maths in English.

Restaurant bills that show a 10% service charge are a good source material for this topic. Learners can be encouraged to see the link between 10% and ÷ 10, which is the same as one tenth.

Learners can bring in promotional leaflets that show percentages and other special offers for use in class. These can be used for price comparisons; for instance, is '50% off' cheaper than 'buy one, get one free'?

The main area of difficulty for many ESOL learners is centred on the use of 'off' and 'of' in exam questions, as they are written similarly, causing learners to get the wrong answer if they do not realise how many 'f's there are. For instance:

Example 1
Find 20% of £40

Example 2
'20% off £40'. What is the new price?

The answer to the first question is £8, but the answer to the last question is £32.

Activity 9

Find 15% off £34 without using a calculator:

10% is 3.4; 5% is half of that, so 1.7.

5% and 10% make 15%; 3.4 + 1.7 = 5.1

So 15% off 34 is 5.1

Answer £5.10

What mistake has the learner made in their answer?

Step 2b

Ratio and Proportion

Ratio

Many languages do not express ratio in the written form that we are familiar with in English, i.e. 1:2. The colon that is used for ratio in English is used as the division sign in much of the rest of the world. This means that learners may initially interpret 1:2 as 'one divided by two', as they are not familiar with the ÷ symbol.

There are many everyday examples of ratios on packaging, such as drinks, paint, and hair dye, which can be brought empty into class to show learners. Learners can experience ratio in the classroom by making up a squash drink, using squash and water. Squash can be measured using measuring spoons or jugs, or a cup.

Learners can be shown that ratios are closely linked to fractions: in a ratio of 1:3, there are 4 parts in total, so one quarter is squash and ¾ is water. Ratios, like fractions, should always be in their lowest form, so 2:6 will be written as 1:3.

Direct Proportion

It can be very important when, for instance, changing a recipe, or mixing cement, to keep the ingredients in the same proportion. Otherwise the recipe or mix will not work. Keeping contents in balance according to a set requirement can involve scaling up or down, depending on circumstances.

A simple recipe for, say, fruit salad for 4 people, can be used as an example for scaling for different numbers of people, whilst keeping the ingredients in proportion.

Proportion can also be used to evaluate prices on supermarket websites. For instance if a 500g bag of peas costs £1, but a 1kg bag costs £1.90, then it is cheaper per pea to buy the larger pack; the prices are not in direct proportion.

Activity 10

Question: The ratio of cats to dogs at a boarding kennels is 1:3. There are twelve dogs at the kennels. How many cats are there?

The learner gives this answer: There are 4 cats.

What error has this learner made, and how could you help correct this misconception?

Metric and Imperial systems of measure

This step will help you to answer the following questions:

- How will most ESOL learners measure length, weight and capacity?
- Which system is used in real life situations in the UK?
- How can ESOL learners be helped to understand the mix of measuring systems currently in use?

During the early part of the 1970s the United Kingdom joined the European Economic Community, or EEC. Prior to this the UK used the imperial system of measures; since then it has used a mix of metric and imperial systems. The majority of the rest of the world only uses the metric system.

The metric system is a decimal system, using base 10. It is built around a one litre cube: this cube measures 10 cm x 10 cm x 10 cm = 1,000 cubic centimetres. When filled with plain water it will hold exactly 1 litre of water (1,000 millilitres), which will weigh 1 kilogram (1,000 grams).

The Imperial system never uses 10 as a base: it uses 8, 12, 14, 16 or 20, depending on which conversion is required.

In the UK lengths became metric first, but the longer length of distance has never been converted from miles to kilometres, so ESOL learners will need to be familiar with this imperial measurement.

Most grocery items are now sold in metric weights, but individuals seem most likely to express their weight in stones and pounds. Also many old cookery books will only show imperial weights for recipes; more recent ones may show metric and imperial, and very recent ones might show only metric.

Fortunately for the haulage industry, a metric tonne and an imperial ton are very similar in weight, with the tonne slightly lighter.

Capacity was the last part attempted in metrification. Many products are sold in millilitres, centilitres and litres, but the imperial equivalent is often still displayed. There are examples of products still sold in imperial measures, such as pints of milk or beer.

Measuring equipment sold in the UK, such as rulers, tapes, jugs and scales usually show both systems, which will help teachers and learners. Learners will also need to be familiar with comparatives and superlatives from the lowest maths levels (E1/N1).

Length and Distance

Practical activities can be the most effective way of introducing learners to an unfamiliar measuring system. For instance many small items in the classroom, or belonging to the learners, can be measured using a ruler, and a chart be drawn to record lengths in each system, metric and imperial.

Learners can then use this information to order the objects in terms of size. The language of length comparison will need to include big, bigger, biggest, long, longer, longest, and short, shorter, shortest, comparing at least two items (E1/N1). Learners will need to know that 'long' can also refer to time.

It may help learners to know their height in feet and inches. Paired activities with learners measuring each other's height using both systems can help make comparative connections between both standard systems (E2/N2).

A fun measuring activity that can take place in the class or at home, is to measure a room for new carpet. Learners can work out how many square metres of carpet will be needed and how many lengths of tracking rod, to hold the carpet to the edges of the room (L1/N4). Non-standard systems can be illustrated by working the amounts out roughly, using strides instead of a tape measure (E2/N2).

Calculating distances between towns, cities or countries that learners are familiar with can help learners to appreciate that miles are longer than kilometres, so they can expect mileage to be a smaller number than the equivalent in kilometres. This knowledge can help when learners are asked to convert between metric and imperial distances (L2/N5).

Map books often contain a distance chart, to show the number of miles between cities in the UK. Learners can see how far it is to other places where they might have family or friends, and convert the distances to kilometres.

Activity 11

When reading the imperial edge of a ruler, what mistake might ESOL learners make when looking at the small marks between the inches?

Step 3a

Weight

Weighing scales are useful in the classroom when teaching weight. If scales are available, asking learners to weigh a number of small objects using firstly the system they are familiar with, and then the 'new' one can help them understand how the two systems work. Record the results on a chart so they can be checked and confirmed.

Weighing scales come in two different forms: those with a dial and arrow, and those which are electronic and have a display of numbers. The latter are usually more accurate. Reading scales accurately to the nearest labelled division is an important life skill that will also be tested in maths exams (E2/N2).

Comparative and superlative language will also be needed, for instance in questions which ask which item is the heaviest, or lightest (E1/N1)? Accurate weighing is not needed for this activity as learners can judge the weight of objects by picking them up.

As the majority of people in the UK prefer to express their weight in stones and pounds learners might like to know theirs too. There is a choice when teaching this: either they can weigh themselves on scales, or they can convert their metric weight to imperial by dividing by 6.5, as there are roughly 6.5 kg in a stone (L2/N5). They may prefer to do this privately, rather than as a shared or group activity.

Activity 12

A learner has converted their weight from metric to imperial by dividing 75kg by 6.5. They have given their answer as 11 stone 5 lbs.

What error has the learner made?

Step 3b

Capacity

Although millilitres and litres have been used for over 30 years in the UK, there are many examples of the old imperial system of fluid ounces, pints and gallons in evidence. For instance milk and beer is still available in pints in the UKt and cars are often quoted as doing a certain number of miles to the gallon.

A 1 litre cube, a collection of different sized and shaped containers, including measuring jugs, bottles and cartons, and some water can be useful when teaching learners about capacity, as it is surprisingly difficult to judge comparative sizes of containers. Alternatively, containers that have the capacity marked on them can be used.

Milk containers, for instance, vary in height and circumference, and a small difference in the latter can make a big difference to the capacity. Thus a two pint container can look smaller than a one litre one, although it contains more milk (about 20%), if the latter is taller. Note the use of comparative language again here.

Examples of non-standard units for measuring capacity that can be shown to learners could include cups, which are still widely used in some countries, including the USA, as a non-sophisticated and cheaper way of measuring. Some measuring jugs also have markers for weights of ingredients, such as flour or rice.

There are roughly 4.5 litres in a gallon. Learners can convert from one to the other by either multiplying or dividing by 4.5.

Using the metric system it is very easy to find the capacity of a 3D shape if given its dimensions, because 1,000 cubic centimetres is one litre. The same cannot be said of the imperial system.

Activity 13

Question: If there are 8 pints in a gallon, and 4.5 litres in a gallon, what is a rough comparison between a pint and a litre?

The learner has given this answer: There are 1.78 litres in a pint.

What error have they made?

Step 3c

In this module you have considered some of the issues around teaching ESOL Maths. Now spend a short time reflecting on what you have learnt.•

Stage 1

Consider what you want to remember about:

- how the numbering system in English is different from other languages
- what challenges learners face when rounding and estimating in maths problems
- how the written form of number operations in English can impact on understanding
- which fractions have non-regular names
- why learners make a mistake when finding, say, 30% off £5
- why ratio and proportion questions place particular demands on ESOL learners
- how most ESOL learners measure length, weight and capacity
- which system is often used in real life situations in the UK
- how ESOL learners can be helped to understand the mix of measuring systems currently used in the UK

Stage 2

Reflect by considering your thoughts and feelings about the content in the module and making some notes.

- What do you feel about the nature of ESOL Maths?
- Have there been any changes in your thinking about the challenges of teaching ESOL Maths, and the potential benefits to learners?

Reflection

- How will what you have learnt affect the people you teach in their everyday lives?

Stage 3

- How can you apply what you have learnt in this module to your own teaching context?
- How will you apply your learning from the module?
- What will you do in the classroom that you didn't do before?

Reflection

Suggested answers to activities

Activity 1

a) There is no u in forty; b) And is inserted after one hundred; c) Five hundred appears twice in the answer

Activity 2

Learners could place a comma between the pounds and pence: £23,68 rather than £23.68

Activity 3

Example: Round £12.49 to the nearest £5

Activity 4

The key word is *approximately*

Activity 5

Sum and total are also nouns: 23 lots of 7 bananas is a sum: 23 x 7. The total number of bananas is 161

Activity 6

Learner could read the first question as 5 – 92

Activity 7

Learner might write £0.6, or £0.60p

Activity 8

Nought point zero, zero, one is one of a number of ways. There is no th consonant cluster in many languages

Suggested answers to activities

Activity 9

Learner has misread off as of so not finished the question: £34 - £5.10 = £28.90

Activity 10

Learner has read the colon as a division sign, so has divided by 3, rather than add 1 and 3, then ÷ 4. There are 3 cats

Activity 11

Learner could assume ten divisions between inches, rather than 16; they could write 1.3 inches, rather than 1 and 3/16ths inches

Activity 12

The answer is 11.5 stone, or 11 ½ stone. Learner has read as 11 stone 5 lbs, rather than working out that half a stone is 7 lbs

Activity 13

Learner may think that a pint is bigger than 1litre, or that 8 ÷ 4.5 gives the number of litres in a pint, rather than pints in a litre. Also note that this is an exact answer, not estimation. A more correct answer is There are roughly two pints in 1 litre

Glossary

Approximation: a result that is not exact but sufficiently close to be useful in a practical context.

Capacity: a volume that can be measured in litres (liquids), or in cubic capacity, such as cubic metres.

Commutative: an operation is commutative if the numbers can change position in the sum without affecting the outcome, as in 3 + 4 = 4 + 3. Addition and multiplication are commutative.

Direct proportion: amounts or quantities that increase or decrease in the same ratio. For instance if two apples cost 50p, and four cost £1, the prices are in proportion: each apple is 25p.

Estimate: arrive at a rough answer by using suitable approximations.

Imperial: a system of measurement used in the UK until the 1970s, when it was partly superseded by the metric system. Includes as units of measure miles, stones and pints, all still in use in the UK.

Metric: a system of measurement designed around the one litre cube. One litre of plain water measures 10 cm x 10 cm x 10 cm, and weighs one kilogram.

Non-commutative: an operation is non commutative if the outcome is affected by changing the position of the numbers in a sum. Subtraction and division are non-commutative.

Non- standard unit: unit of measure which can vary, such as a cup or a stride.

Product: in mathematics another word for multiply. Has other definitions.

Ratio: a way of comparing the amounts or quantities that are in a specific relationship with each other, i.e. two to one. Written 2:1.

Round: to adjust a number to a specified level. For instance, 17.9 to the nearest 10 is 20.

Sign: in mathematics the generic word for the symbols +, -, x and ÷.

Standard unit: a unit of measure in an official system, such as metric or imperial.

Glossary

Further reading

Adult Numeracy Core Curriculum: EXCELLENCE GATEWAY (2011). *SFL Curriculum*. [online]. http://rwp.excellencegateway.org.uk/Numeracy/Adult Numeracy Core Curriculum/

ADLER, Jill (2001). *Teaching Mathematics in Multilingual Classrooms*. vol.26. 1st ed., Dordrecht, Neths, Kluwer Academic Publishers. Mathematics Education Library, 26.

BARWELL, Richard (ed.). *Multilingualism in Mathematics Classrooms: Global Perspectives*. 2009 1st ed., Bristol, UK.

KERSAINT, Gladis, THOMPSON, Denisse R. and PETKOVA, Mariana (2013). *Teaching Mathematics to English Language Learners*. 2nd ed., Abingdon, Oxon, Routledge.

NCETM (2010). *Top tips for ESOL learners*: [online]. https://www.ncetm.org.uk/public/files/340591/Action_Research_Project_Leicestershire_ALS_Top_Tips.pdf

NEWMARCH, Barbara (2005). *Developing Numeracy, Supporting Achievement*. Leicester, NIACE. Lifelines in Adult Learning, 19.

OFQUAL (2010). *Fair Access by Design*. Coventry. Guidance Document 040/2010.

STACEY, Jennifer M (2013). *Teaching and Learning in ESOL Maths*. [online]. At www.esolmaths.co.uk

SWAN, Michael and SMITH, Bernard (eds.) (2001). *Learner English- A teachers guide to interference and other problems*. 2nd ed., Cambridge, C.U.P.

WEAVER, Janine (2010). *ESOL learners don't understand the questions- what can we do?* [online].